

no retraining,
no signal labels

Calorimeter shower



$$x = (E_0, E_1, E_2, \hat{I})$$

photon shower,
or $\chi \rightarrow \gamma\gamma$

Trained CaloFlow ❄️

weights frozen

$$\text{Flow-I: } p_I(E_0, E_1, E_2)$$

$$\text{Flow-II: } p_{II}(\hat{I} | E_{0..2})$$

$$\log p = \log p_I + \log p_{II}$$

Flow-II latent

$$z_{II} \in \mathbb{R}^{504}$$



8 architecture-fixed
regions G

Per-region excess

$$s_G = \frac{\sum_{a \in G} z_a^2 - |G|}{\sqrt{2|G|}}$$

$$M = \max_G s_G$$



Mondrian split-conformal

calibrated p -value



finite-sample, per-event
false-positive control

inherited (Krause et al.)

this work: reuse the latent, score, calibrate